

ReserveChain

Industrial-Metals Reserve Registry and Tokenization Framework

INSTITUTIONAL WHITEPAPER · DISCUSSION DRAFT · IN DEVELOPMENT

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reservechain.io

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EU/EEA notice. ReserveChain does not currently intend to offer tokens to residents of, or persons located in, the EU/EEA.

Important Notice

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1. Abstract

ReserveChain is an in-development platform designed to register, evidence and — only if and when legally authorised — represent on a blockchain specific physical lots of two industrial metals: **ultra-high-purity copper powder** (program identity **Cu 29**) and **high-purity nickel wire** (program identity **Ni 28**).

The platform's central proposition is that a token representing a physical asset is only as credible as the evidence behind it. ReserveChain therefore puts the **evidence layer first**: an asset registry of lots, batches, containers and coils; a library of certificates of analysis, custody records, insurance documents, valuations and reserve reports, each fingerprinted with SHA-256; a **Digital Asset Passport (DAP)** for every registered physical unit, with a lifecycle timeline, an evidence ledger and a Merkle root summarising all supporting documents; and a **tamper-evident audit trail** whose state can be anchored on a public blockchain.

On top of that evidence layer, ReserveChain has designed a **token architecture** based on audited open-source building blocks (OpenZeppelin Contracts v5) with compliance-gated transfers, a reserve-gated mint ceiling that is fail-closed by default, a proposed redemption workflow, a role-controlled treasury and an audit-anchoring contract. All economic parameters — supply, asset-to-token ratio, price, redemption thresholds, allocations — are **unset** and remain **to be determined, subject to written approval**. The token, wallet, purchase, Proof-of-Reserves and redemption features are implemented as **inactive modules**, switched off until authorised.

Every factual claim on the platform carries a visible **Claim Status** — *proposed, in development, pending verification, verified* or *not applicable* — so that readers can distinguish intention from evidence at a glance.

Since the previous draft, the platform in development has gained: a **Proof-of-Reserves reconciliation engine** with explicit exception rules and four-eyes-approved, fingerprinted snapshots (Section 14); an off-chain **redemption workflow** with compliance re-checks and unit-level status (Section 13); **wallet linking** and **USDT payment verification** modules that are implemented but **inactive** and restricted to test networks (Section 15); a **participant portal** with audience-restricted data rooms and an **asset-owner intake** pipeline (Section 16); and **operations monitoring, backups and disaster-recovery** procedures (Section 17). For each metal program, the asset owner has supplied a laboratory **Certificate of Analysis**; these are recorded as **owner-supplied** and remain **subject to documentary verification** (Section 4).

2. The Problem: Trust in Physical-Asset Representations

Digital representations of physical commodities promise operational benefits: fractional record-keeping, faster transfer of entitlements, programmable compliance and continuous transparency. In practice, many such representations have suffered from recurring weaknesses:

1. **Opaque backing.** Statements that tokens are "backed" by physical assets are often supported by infrequent, high-level attestations that do not identify specific lots, specifications, locations or custodians.
2. **Claim inflation.** Marketing materials frequently describe arrangements (insurance, audited reserves, redemption rights) as existing when they are only planned.
3. **Unverifiable documents.** Certificates and reports are published as images or PDFs without any mechanism for a reader to confirm that the document they hold is the one the issuer relied on.
4. **Mutable records.** Off-chain databases can be silently altered; on-chain tokens can be minted without reference to off-chain reality.
5. **Weak separation of duties.** The same individual can often draft, approve and publish material claims.

Industrial metals add further complexity. Unlike investment-grade bullion, materials such as metal powders and fine wire are defined by technical specifications (chemistry, purity, particle size or diameter, packaging and handling conditions) that matter to their industrial value and must be evidenced by competent laboratories.

ReserveChain's design responds to each weakness with a specific control, summarised below and detailed in later sections.

Weakness	ReserveChain proposed control
Opaque backing	Unit-level registry and Digital Asset Passports
Claim inflation	Claim Status System; placeholders instead of assumptions; four-eyes approval
Unverifiable documents	SHA-256 fingerprints; client-side verification; Merkle roots
Mutable records	Append-only, hash-chained audit trail with database triggers and on-chain anchoring
Mint without reality	Reserve-gated mint ceiling, fail-closed by default
Weak separation of duties	Role-based access, four-eyes workflow, multisig administration (intended)

3. Project Model

3.1 Overview

ReserveChain is conceived as a **turnkey registry and tokenization platform** with three layers:

- **Evidence layer** (off-chain, system of record): registry, documents, passports, audit trail.
- **Control layer**: editorial workflow, roles, compliance controls, site modes and module flags.
- **Representation layer** (on-chain, testnet in this phase): token, compliance registry, reserve guard, redemption manager, treasury and audit anchor.

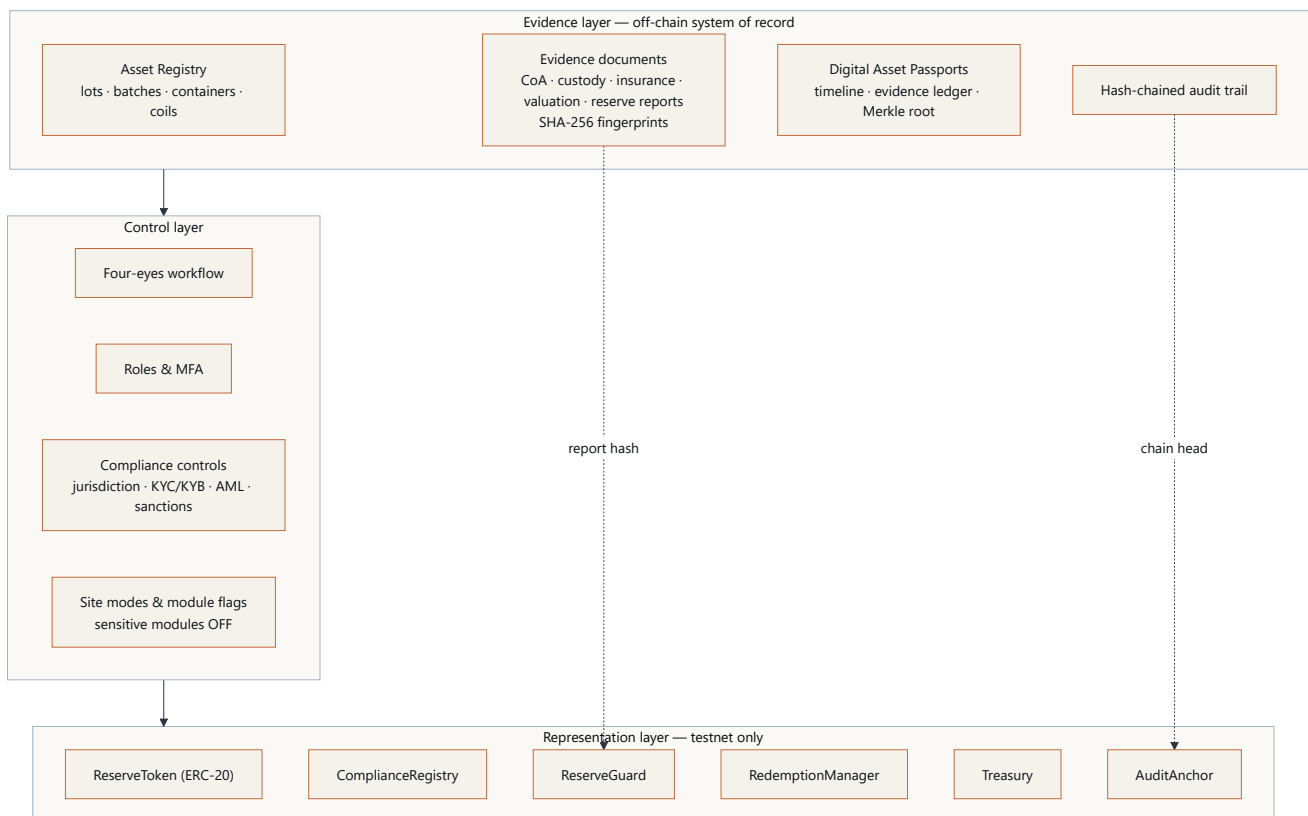


Figure 1 — Platform overview

3.2 Participants (proposed)

Participant	Role	Status
ReserveChain operating entity	Platform operator, registry owner	Proposed Swiss structure — Pending — subject to final approval
Producers / refiners	Supply of material	[To be provided by ReserveChain]
Independent laboratories	Certificates of analysis	[To be provided by ReserveChain]
Custodian(s) / warehouse operator(s)	Safekeeping, receipts	[To be provided by ReserveChain]

Participant	Role	Status
Insurer(s)	Coverage of stored material	[To be provided by ReserveChain]
Independent attester / auditor	Reserve reporting	[To be appointed by ReserveChain]
KYC/KYB/AML/sanctions provider	Eligibility screening	[To be decided by ReserveChain]
Multisig signers	On-chain administration	[To be provided by ReserveChain]
Eligible participants	Prospective token holders	Not applicable — no offering exists

3.3 Proposed lifecycle

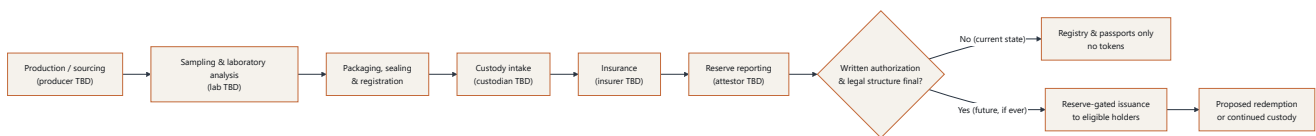


Figure 2 — Asset lifecycle

At present ReserveChain operates only in the state labelled *"Registry & passports only — no tokens"*.

4. The Metal Programs

Provisional Asset Notice. Preliminary illustrative information only - subject to documentary verification, independent assessment and final approval. No asset or token is currently offered for sale through this website.

This section describes the two materials in general industrial terms and reproduces facts stated on **owner-supplied** Certificates of Analysis (CoA). Those facts are recorded as *pending verification*: ReserveChain has **not yet independently confirmed** the certificates with the issuing laboratory, and their reproduction here does not imply any verification, valuation, custody arrangement, reserve claim or token. **No price, value, custody location or counterparty arrangement for ReserveChain's assets is stated or implied.** Further asset-specific data will be published in the registry and Digital Asset Passports only when provided and evidenced.

4.1 Copper Powder — Cu 29

General industrial context. Copper is valued for its high electrical and thermal conductivity. In powder form, copper is used across a range of industrial processes, including powder metallurgy (pressed and sintered components), additive manufacturing, electrically and thermally conductive pastes and inks, brazing materials, friction materials, and electronics and thermal-management applications. Performance in these uses typically depends on chemical purity, oxygen content, particle size distribution, particle morphology, apparent density and flowability, and on handling conditions that limit oxidation.

ReserveChain program parameters.

Parameter	Value	Claim status
Material form	Ultrafine copper powder (as described on owner-supplied CoA 0004512)	Pending verification
Stated purity	99.9999 % as stated on owner-supplied CoA 0004512 — subject to documentary verification	Pending verification
Purity basis	Chemical purity based on the impurities Al, Cd, Fe, Mg, Mo, Ni, Sb, Ti, Zn per TU 1793-011-50316079-2004 (as stated on the certificate)	Pending verification
Particle size distribution	Pending — laboratory analysis required	Pending verification
Packaging	Glass ampoules packed in cardboard boxes (per owner-supplied CoA)	Pending verification
Producer / origin	Pending — producer to be disclosed	Proposed
Declared quantity	Lot #03-K-07: 2000 kg declared (net weight according to data supplied by the customer to the laboratory)	Pending verification
Unit of account	Not yet determined	Proposed

Parameter	Value	Claim status
Laboratory	IGAS research, Goslar — named on the owner-supplied certificate; not yet independently confirmed; no partnership, endorsement or accreditation implied	Pending verification
Custodian / location	Custodian to be appointed — subject to final approval	Proposed
Insurance	Insurance not yet arranged — subject to final approval	Proposed

4.2 Nickel Wire — Ni 28

General industrial context. Nickel is used for its corrosion resistance, its behaviour at elevated temperatures and its electrical and magnetic properties. High-purity nickel wire is used in applications such as electronic and electrical components, lead wires, battery connections, heating and resistance elements (often in alloy form), vacuum and lighting technology, chemical-processing equipment, and as feedstock for specialised alloys and thermal spray. Relevant characteristics typically include chemical purity, diameter and tolerance, temper/annealing condition, surface finish, tensile properties and spool or coil format.

ReserveChain program parameters.

Parameter	Value	Claim status
Material form	Nickel wire, 0.025 mm diameter (as described on owner-supplied CoA 0004368 — dimensional inspection pending)	Pending verification
Stated purity	99.9807 % as stated on owner-supplied CoA 0004368 — subject to documentary verification	Pending verification
Purity basis	Impurities per GOST 2179-75 (As, Cu, Fe, Mn, Pb, Si) stated as 0,0193 % by weight	Pending verification
Wire diameter tolerance	Pending — laboratory analysis required	Pending verification
Temper / condition	[To be provided by ReserveChain]	Proposed
Coil / spool format	30 bobbins in one box (per owner-supplied CoA)	Pending verification
Producer / origin	Pending — producer to be disclosed	Proposed
Declared quantity	Lot 120/NP1: 5000 g total net weight declared (according to information given by the customer to the laboratory)	Pending verification
Laboratory	IGAS research, Goslar — named on the owner-supplied certificate; not yet independently confirmed; no partnership, endorsement or accreditation implied	Pending verification
Custodian / location	Custodian to be appointed — subject to final approval	Proposed

Parameter	Value	Claim status
Insurance	Insurance not yet arranged — subject to final approval	Proposed

4.3 Owner-supplied Certificates of Analysis

The asset owner has supplied one IGAS Certificate of Analysis for each program. ReserveChain has registered a scan of each certificate as an evidence document (SHA-256 fingerprinted; the customer address block is redacted on the supplied copy) and transcribed its stated values into a CoA record. Both records carry provenance **owner-supplied** and Claim Status **pending verification**. The figures below are reproduced as stated on the certificates; they are **not** ReserveChain findings.

Item	Copper — CoA 0004512	Nickel — CoA 0004368
Provenance	Owner-supplied — subject to documentary verification	Owner-supplied — subject to documentary verification
Laboratory named on certificate	IGAS research, Goslar (DE)	IGAS research, Goslar (DE)
Date of certificate	04.07.2022	19.10.2021
Goods described	Ultrafine Copper Powder, Lot #03-K-07	Nickel wire 0,025 mm dia, DKRNT NP1, Lot 120/NP1
Declared quantity	2000 kg (net weight according to data supplied by customer)	5000 g total net weight, 30 bobbins in 1 box (according to information given by customer)
Sample (as stated)	10 g, from Box no. 20, sampled 01.07.2022	0,8 g, from bobbins no. 8, 10, 19 and 27, sampled 14.10.2021
Method	ICP/OES	ICP/MS and ICP/OES
Stated purity	99.9999 %	99.9807 %
Purity basis	Impurities Al, Cd, Fe, Mg, Mo, Ni, Sb, Ti, Zn per TU 1793-011-50316079-2004	Impurities per GOST 2179-75 (As, Cu, Fe, Mn, Pb, Si): 0,0193 % by weight
Verification status	Pending verification — not yet confirmed with the issuing laboratory	Pending verification — not yet confirmed with the issuing laboratory

Important limitations:

- The declared quantities are **net weights declared by the customer** to the laboratory; they have not been weighed or verified by ReserveChain, by a custodian or by an independent inspector.
- A certificate describes the sample analysed at the time stated. It does not establish the present existence, location, condition, ownership or quantity of the material.
- A purity figure computed on a stated impurity basis is not necessarily comparable with purity figures computed on other bases or standards.
- Physical-property and dimensional fields not covered by the certificates (for example particle size distribution, diameter tolerance and temper) remain **pending**.

- In the Proof-of-Reserves reconciliation (Section 14), a unit whose only certificate is owner-supplied raises an *owner-supplied only* exception until independent verification is recorded.

4.4 Why unit-level registration matters for industrial metals

Industrial materials are not fungible in the way that standardised bullion bars are. Two lots of nominally identical copper powder may differ in particle size distribution or oxygen content, and two coils of nickel wire may differ in temper or diameter tolerance. ReserveChain's proposed registry therefore tracks **specific physical units** — lots, batches, containers and coils — each with its own evidence and passport, rather than an undifferentiated pool. Any future token program would need to define, in its offering documentation, exactly which registered units it relates to and how differences between units are treated. That definition is **[To be determined — subject to written approval]**.

5. Proposed Legal, Custody and Reserve Frameworks

All arrangements in this section are **proposed**. None is in place. Their final form depends on legal advice, regulatory analysis and commercial agreements that have not been concluded.

5.1 Corporate and legal structure

ReserveChain intends to operate through a **proposed Swiss structure**. The legal form, registered office, governing bodies, licensing analysis and any required registrations are **Pending — subject to final approval**. References elsewhere to other jurisdictions (including Estonia) are for context only; no entity is represented as formed in any jurisdiction.

Key legal questions to be resolved by ReserveChain's counsel before any token issuance include:

Question	Status
Legal characterisation of any token (e.g. ledger-based security, payment token, utility, other) under applicable law	Pending — subject to legal analysis
Holder rights (ownership, claim, entitlement to delivery, none)	Not yet determined — subject to final legal structure
Bankruptcy remoteness / segregation of physical assets	Pending — subject to legal analysis
Licensing or registration requirements (including AML self-regulatory organisation membership, if applicable)	Pending — subject to legal analysis
Offering documentation and investor categories	Pending — subject to final approval
Target jurisdictions and exclusions (EU/EEA currently excluded by intention)	Pending — subject to final approval
Tax treatment	Pending — subject to advice

5.2 Custody framework (proposed)

The proposed custody model aims to give each registered physical unit an evidenced chain of custody:

1. **Custody intake** — receipt of sealed units by a custodian or warehouse operator, with a receipt identifying unit IDs and seal numbers.
2. **Segregation** — physical and book-entry segregation of program material from other holdings (type of segregation **[To be determined]**).
3. **Periodic inspection** — counts and seal checks by the custodian and, where agreed, by an independent inspector.
4. **Transfers and releases** — every movement recorded as a custody record linked to the unit's passport.

5. Environmental controls — storage conditions appropriate to the material (e.g. humidity and oxidation control for fine powders), to be specified with the custodian.

Element	Value
Custodian(s)	[To be provided by ReserveChain]
Facility location(s)	[To be provided by ReserveChain] — public disclosure level to be decided
Custody agreement	Pending — subject to final approval
Warehouse receipt form	[To be provided by ReserveChain]
Legal owner of the material	Pending — subject to final legal structure

5.3 Insurance framework (proposed)

ReserveChain intends to seek insurance appropriate to stored industrial materials (e.g. property/all-risk while in custody, and transit cover during movements). Insurer, coverage type, sum insured, exclusions and policy period are [To be provided by ReserveChain]. Until a certificate of insurance is registered and verified, the platform displays *"Insurance not yet arranged — subject to final approval."*

5.4 Reserve framework (proposed)

The proposed reserve framework links three items for each program:

- **Registered units** — the set of Published registry units (with passports) designated to the program;
- **Reserve reports** — periodic statements by an independent attestor of the units held, with the report document fingerprinted;
- **Outstanding representation** — tokens outstanding, if any (currently: none).

Any future issuance would be limited by the reserve-gated mint ceiling (Section 7.4). The attestor, reporting frequency, procedures (e.g. agreed-upon procedures vs. assurance engagement) and acceptance criteria are [To be determined — subject to written approval].

6. Digital Asset Passport and Evidence Model

6.1 Registry hierarchy

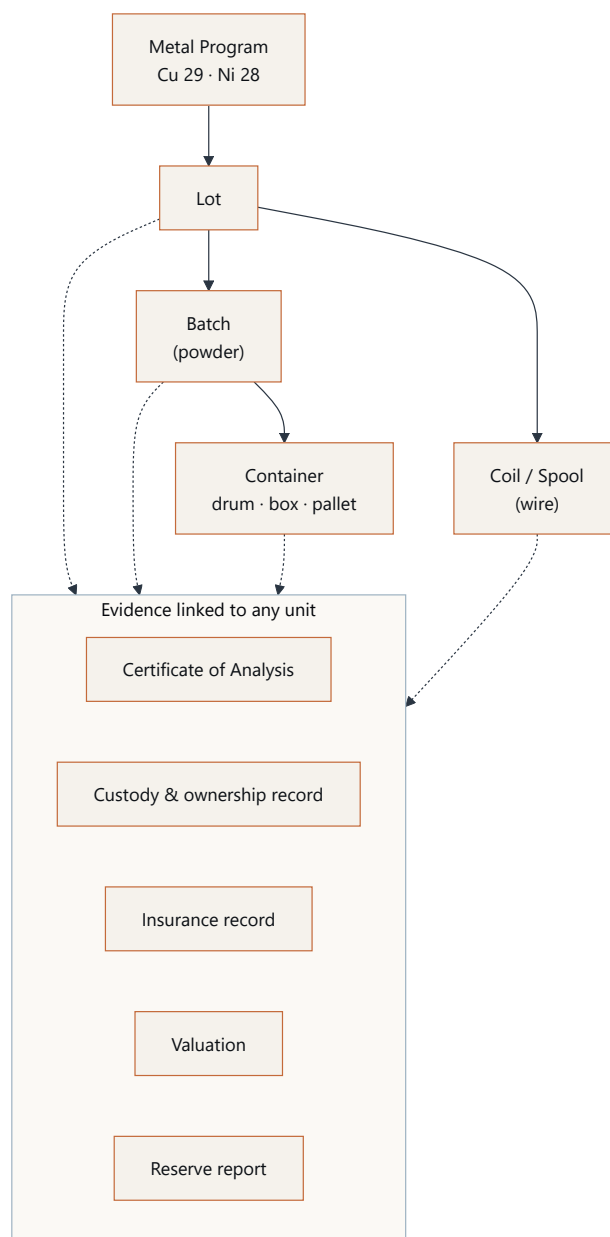


Figure 3 — Registry hierarchy

Every registry record carries a **verification status** (Claim Status) and a **workflow state**. Only records that have passed four-eyes review and are in the *Published* state are visible publicly.

6.2 The Digital Asset Passport

A Digital Asset Passport is generated for every Published lot, batch, container and coil. It contains:

Section	Content
Identity	Passport number, entity type, program, registry number, QR code
Fields	Every public attribute with its value or its explicit pending text, plus a Claim Status per field
Completeness	Share of public fields that have evidenced values — gaps are displayed, not hidden
Timeline	Lifecycle events derived from evidence (analysis, intake, transfers, reports)
Evidence ledger	Linked documents with type, issuer (or "Not yet provided"), status and SHA-256 fingerprint
Custody	Custody and ownership records
Merkle root	A single SHA-256 commitment over all evidence fingerprints
Anchoring	Reference to on-chain anchor, where enabled

6.3 Fingerprinting and Merkle commitment

Each evidence document is hashed with SHA-256 at ingest. The hash is stored with the document record and becomes immutable once the document is Published. For each passport, the sorted list of evidence fingerprints forms the leaves of a Merkle tree; its root is displayed on the passport.

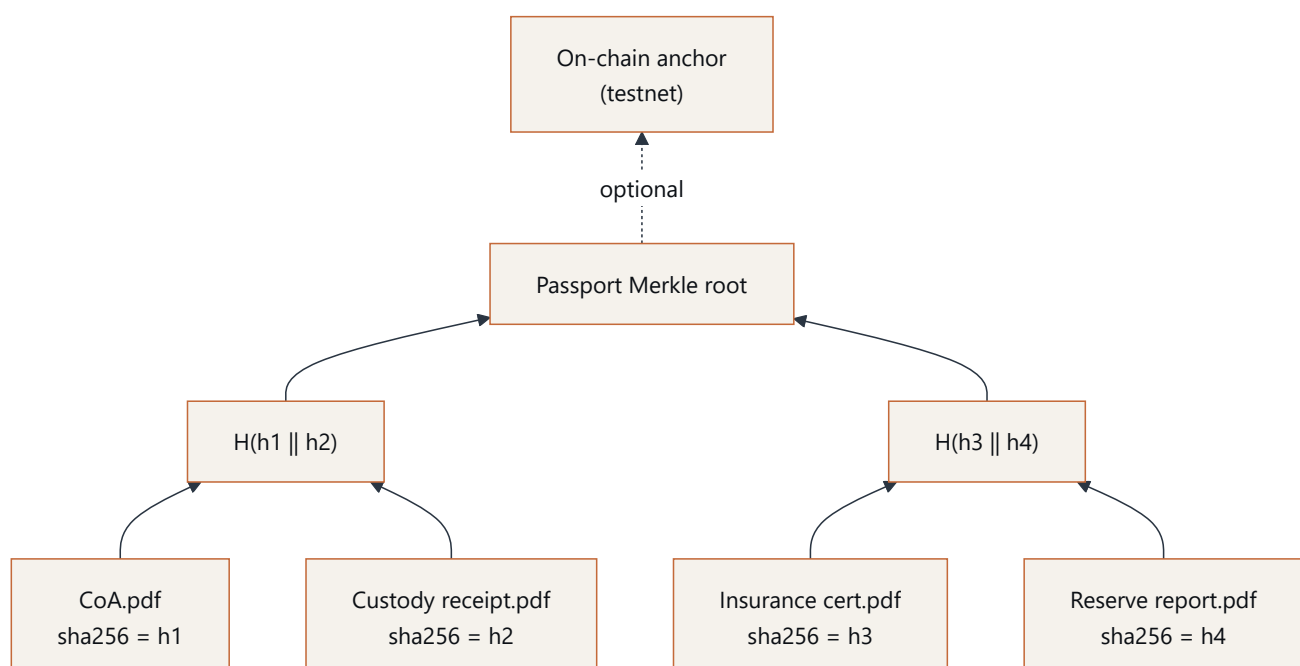


Figure 4 — Evidence merkle

Properties:

- Changing, adding or removing any evidence document changes the passport's Merkle root.
- A third party holding a document can prove its inclusion with a short Merkle proof without the issuer disclosing other documents.
- Anchoring a root (or the audit-trail head that records it) on a public chain fixes it in time.

6.4 Client-side verification

Any visitor can verify a document they hold: the file is hashed **locally in the browser** (or app) using the Web Crypto API; only the 64-character hash is sent to the registry's verification endpoint, which responds whether the fingerprint matches a registered document and, if so, which passport it belongs to. The file itself is never uploaded.

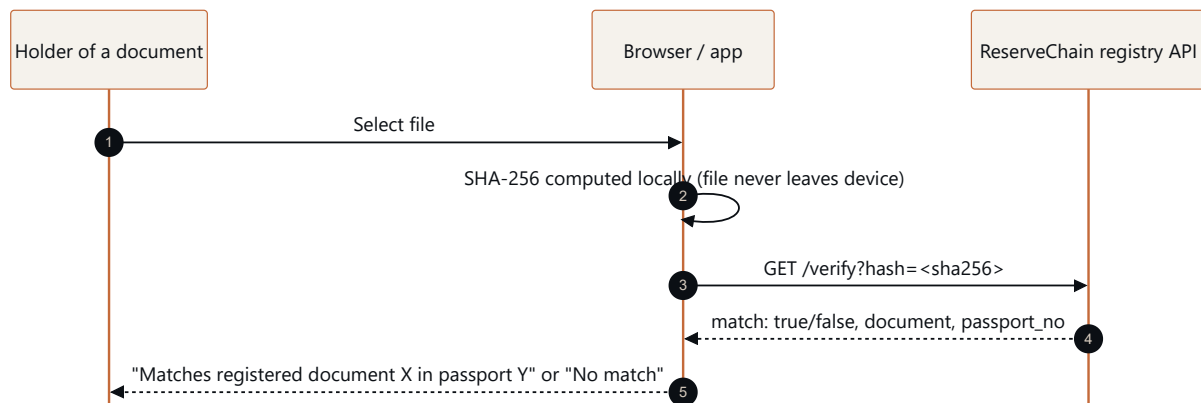


Figure 5 — Client verify

6.5 Claim Status System

Status	Meaning	Who may set
Proposed	Intended arrangement, not yet in progress	Editors / registry managers
In development	Work under way, not complete	Editors / registry managers
Pending verification	Evidence submitted, not yet verified	Registry managers
Verified	Evidence reviewed and accepted under four-eyes control	Compliance approval only
Not applicable	Does not apply in the current phase	Editors with review

7. Token Architecture

Contracts described here exist as source code with automated tests and are intended for **test networks only** (Ethereum Sepolia by default; Polygon Amoy optional). **Mainnet deployment is not performed.** No token has been issued to any person.

7.1 Contract suite

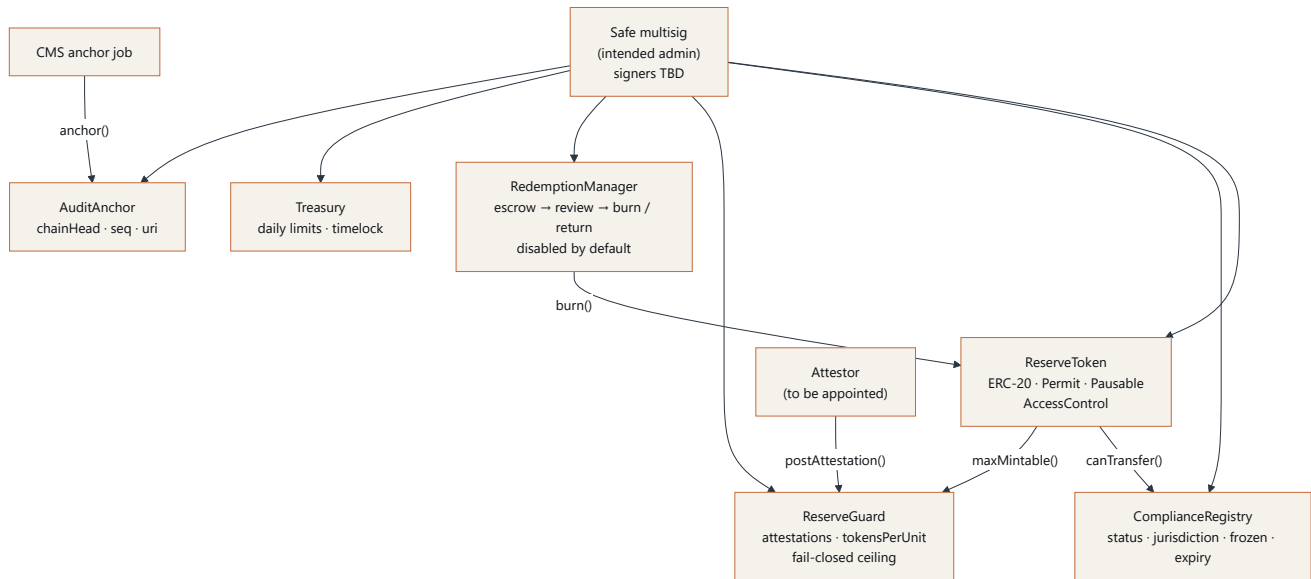


Figure 6 — Token architecture

Contract	Purpose	Key properties
ReserveToken	ERC-20 representation (if ever authorised)	OpenZeppelin v5; ERC20Permit; Pausable; roles MINTER, BURNER, PAUSER, COMPLIANCE_ADMIN, TREASURY, DEFAULT_ADMIN; name, symbol, decimals and supply cap from configuration; optional compliance hook and reserve guard; document URI + hash reference
ComplianceRegistry	On-chain eligibility flags	Per-address status, two-letter jurisdiction code, frozen flag, expiry; blocked-jurisdiction list; roles KYC_OPERATOR and JURISDICTION_ADMIN; no personal data on-chain
ReserveGuard	Reserve-gated mint ceiling	Attestor posts <code>(programId, units, reportHash, uri, asOf)</code> ; ceiling = units × tokensPerUnit; staleness window; returns zero unless fully configured
RedemptionManager	Proposed redemption workflow	Request escrows tokens; operator approves (burn) or rejects (return); min/max thresholds; disabled by default
Treasury	Role-controlled holdings	Per-token daily limits; optional timelocked withdrawals; pausable

Contract	Purpose	Key properties
AuditAnchor	Anchors the CMS audit-trail head	Monotonic sequence numbers; <code>verify(seq, chainHead)</code> ; <code>ANCHOR_ROLE</code> only

7.2 Roles and separation of duties

Role	Contract	Intended holder
DEFAULT_ADMIN (with delayed transfer rules)	All	Safe multisig — signers and threshold [To be provided by ReserveChain]
MINTER	ReserveToken	Multisig; minting additionally bounded by ReserveGuard
BURNER	ReserveToken	RedemptionManager / multisig
PAUSER	ReserveToken, RedemptionManager, Treasury	Multisig and optional emergency signer
COMPLIANCE_ADMIN	ReserveToken	Compliance multisig
KYC_OPERATOR, JURISDICTION_ADMIN	ComplianceRegistry	Compliance operations
ATTESTOR	ReserveGuard	Independent attester [To be appointed]
GUARD_ADMIN	ReserveGuard	Multisig
REDEMPTION_OPERATOR, CONFIG_ADMIN	RedemptionManager	Operations / multisig
TREASURER, LIMIT_ADMIN	Treasury	Multisig
ANCHOR_ROLE	AuditAnchor	Platform anchor key (low privilege)

7.3 Compliance-gated transfers

When compliance is enabled on the token, every balance movement — transfers and the recipient side of minting — calls `ComplianceRegistry.canTransfer(from, to, amount)`. A movement is permitted only if the parties are eligible: approved status, not frozen, record not expired, and jurisdiction not on the blocked list. EU/EEA jurisdiction codes are proposed to be blocked by default in line with the EU/EEA notice above. All movements are also blocked while the token is paused.

7.4 Reserve-gated minting (fail-closed)

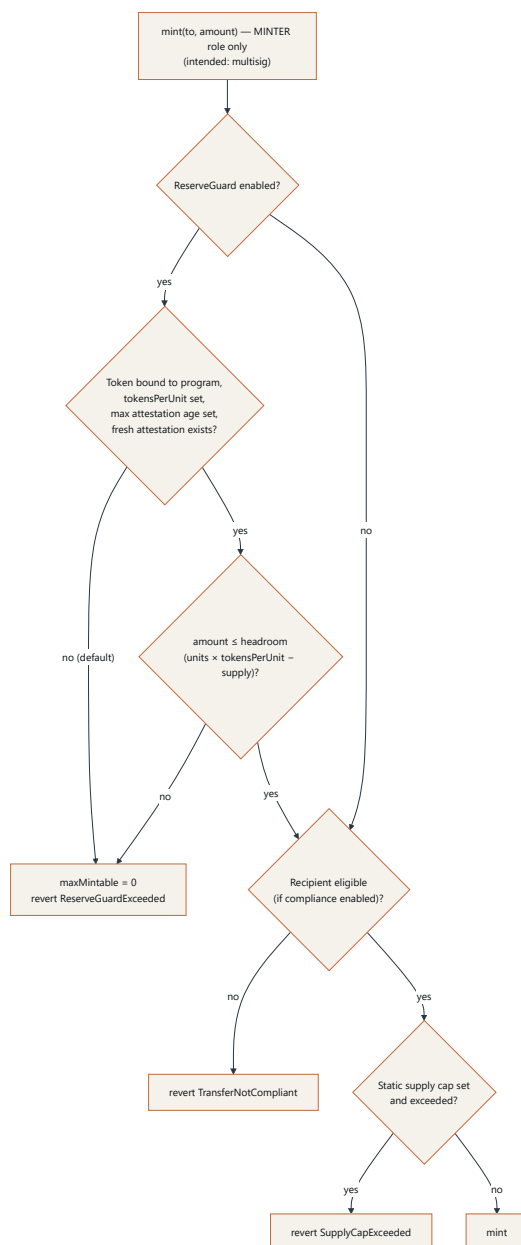


Figure 7 — Reserve gated mint

When the ReserveGuard is enabled, its ceiling is zero unless the token is bound to a program, the ratio (`tokensPerUnit`) is set, the staleness window is set and a fresh attestation exists — all of which are **unset by default**. Minting is therefore impossible until each element has been deliberately configured. The intended operating policy is that any token representing registered material is deployed with the ReserveGuard enabled, that the MINTER role is held only by the multisig, and that no minting takes place during the current phase. The reserve ceiling is a technical safeguard; it is **not** itself a Proof of Reserves, which depends on the independence and quality of the attestation behind each posted figure.

7.5 Configuration, not code

Token name, symbol, decimals, supply cap, ratio, thresholds and limits are supplied through per-network configuration files and administrative transactions. Nothing economic is hard-coded. Default

configuration leaves every economic parameter unset.

8. Tokenomics Framework

No tokenomics have been decided. The table below is a *framework* listing the parameters that would need to be defined, approved in writing and disclosed in definitive offering documentation before any issuance. Every value is **To be determined — subject to written approval.**

Parameter	Description	Value
Token name and symbol	Identifier(s) per program	To be determined — subject to written approval
Decimals	Divisibility	To be determined — subject to written approval
Unit of account	Physical unit underlying the representation (e.g. mass unit)	To be determined — subject to written approval
Asset-to-token ratio	Tokens per unit of registered material	To be determined — subject to written approval
Supply cap	Maximum tokens per program	To be determined — subject to written approval
Issuance price / pricing methodology	How any issuance price would be set	To be determined — subject to written approval
Allocations	Any allocation categories	To be determined — subject to written approval
Holder rights	Legal nature of any entitlement	To be determined — subject to final legal structure
Redemption minimum / maximum	Thresholds per request	To be determined — subject to written approval
Redemption fees and delivery terms	Costs and logistics	To be determined — subject to written approval
Custody and insurance cost allocation	How ongoing costs are borne	To be determined — subject to written approval
Transfer restrictions	Eligibility rules, lock-ups	To be determined — subject to written approval
Secondary transfer venues / liquidity	Any trading arrangements	To be determined — no liquidity is represented
Reporting frequency	Reserve report cadence	To be determined — subject to written approval

ReserveChain makes **no statement** regarding price, value, appreciation, liquidity, yield or return of any future token.

9. Governance

9.1 Principles

1. **Separation of duties** — no single person can author, approve and publish a material claim.
2. **Least privilege** — role-based access in the CMS and on-chain.
3. **Traceability** — every material action is recorded in a tamper-evident audit trail.
4. **Explicit authorization** — sensitive modules require a written authorization reference and two distinct authorised users.
5. **External verifiability** — fingerprints, Merkle roots and audit anchors allow outside parties to check integrity.

9.2 Editorial and registry workflow

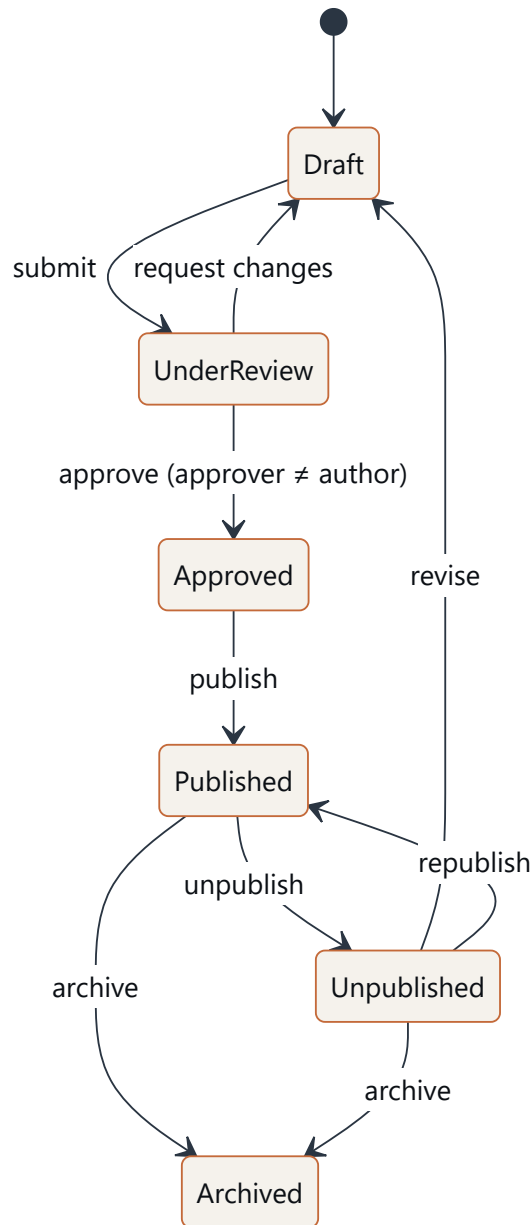


Figure 8 — Four eyes workflow

9.3 Platform roles

Role	Responsibility
Content editor	Drafts public content; submits for review
Registry manager	Creates and maintains registry and evidence records; submits for review
Reviewer	Reviews and approves others' submissions
Compliance officer	Publishes; approves compliance-sensitive changes; manages jurisdictions, eligibility statuses and waitlist data
Auditor	Read-only access; verifies audit-trail integrity; exports evidence
Administrator	Platform settings, module authorization (with written reference), audit anchoring

9.4 On-chain governance (intended)

Administrative rights over contracts are intended to be held by a **Safe multisig** whose signers and threshold are **[To be provided by ReserveChain]**. Admin transfers use delayed two-step rules. Emergency pause is available to designated pausers. All role changes are to be mirrored in the CMS audit trail with transaction references.

9.5 Corporate governance

Board composition, committees (e.g. risk, compliance), conflicts-of-interest policy and external audit arrangements of the proposed entity are **Pending — subject to final approval**.

10. Technology Platform

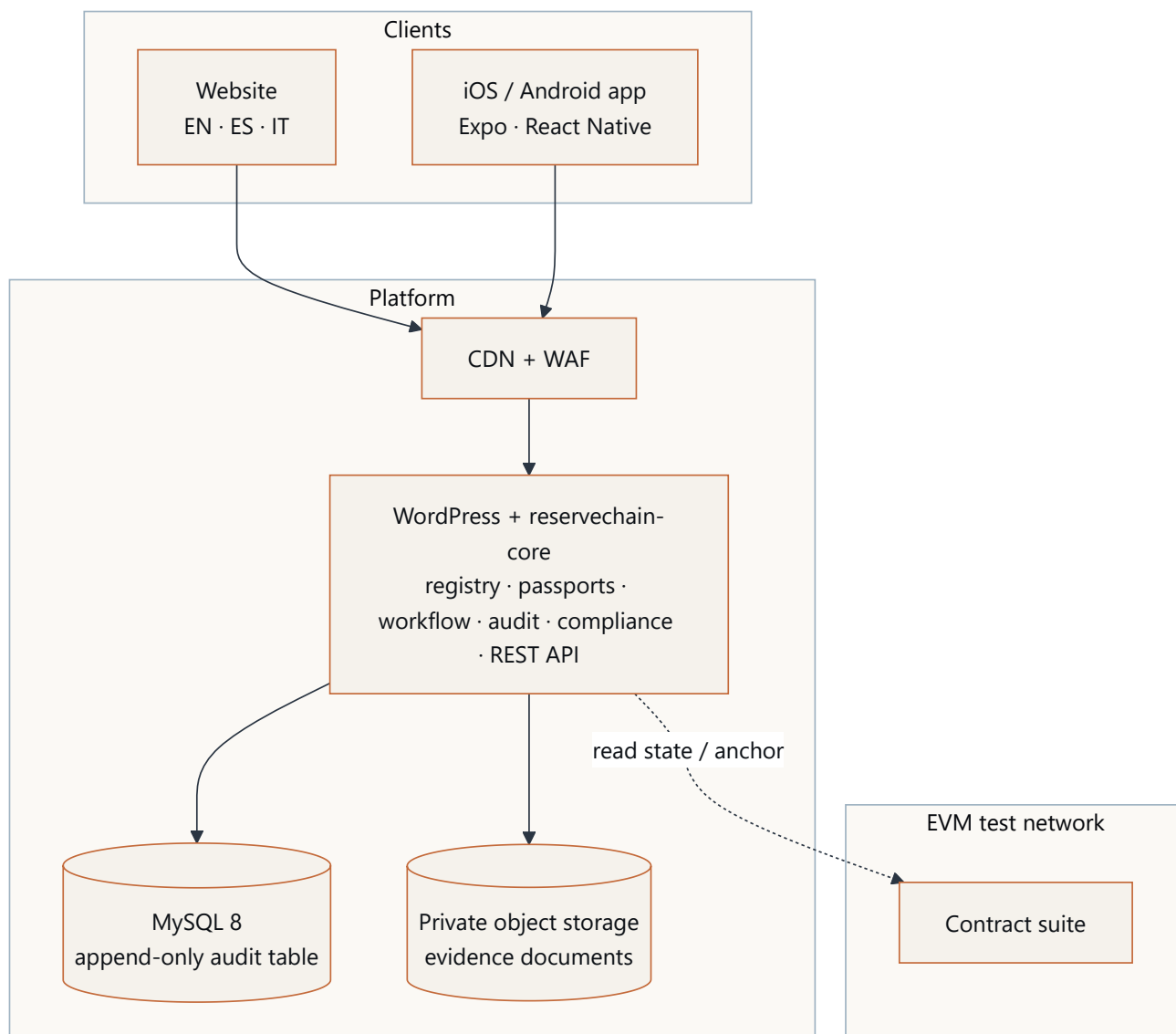


Figure 9 — Technology stack

Component	Technology	Notes
Website and CMS	WordPress, custom theme and plugin (PHP 8.1+), MySQL 8	Declarative registry schema; extensible without rebuild
REST API	<code>/wp-json/rc/v1</code>	Public read-only endpoints; authenticated member endpoints with MFA
Mobile apps	Expo / React Native (TypeScript), native builds for iOS and Android	Remote configuration; inactive modules hidden
Smart contracts	Solidity, Hardhat, OpenZeppelin Contracts v5	Testnet only
Languages	English, Spanish, Italian	Legal texts subject to counsel-approved translation

Component	Technology	Notes
Environments	Development, Staging, Production — strictly separated	Separate data, secrets and access

Audit trail. Each audit entry stores the hash of the previous entry and its own SHA-256 hash over a canonical field sequence. Database triggers reject any UPDATE or DELETE on the audit table. An integrity tool recomputes the entire chain on demand and daily. The latest chain head can be anchored to the AuditAnchor contract; any later rewrite of history becomes detectable against the anchor.

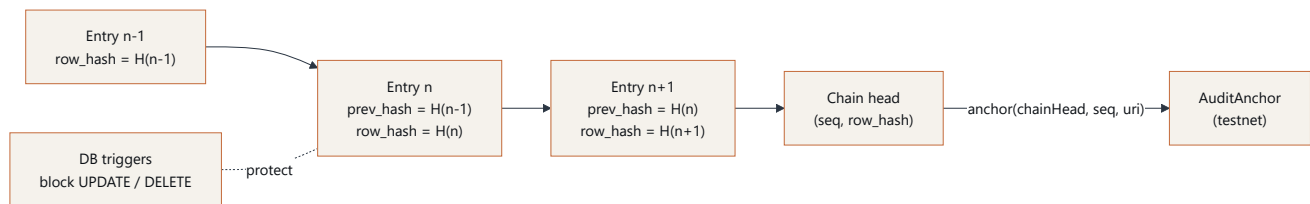


Figure 10 — Audit anchoring

11. Security

ReserveChain's proposed security programme comprises:

Domain	Proposed controls
Identity	Named accounts; role-based access; mandatory TOTP multi-factor authentication for staff
Sessions	Secure, HttpOnly cookies; idle and absolute timeouts; revocation on role change
Transport & browser	TLS 1.2+, HSTS, Content Security Policy, hardened headers
Application	Prepared statements, output escaping, nonces, capability checks; file editing disabled in production
Uploads	Type allow-list, magic-byte validation, malware scanning, private storage, signed URLs, SHA-256 at ingest
Abuse	Edge WAF, rate limiting, honeypots, optional challenge
Secrets	Secrets manager; no secrets in source; scheduled rotation
Data protection	Hashed IPs and emails for de-duplication; PII restricted to compliance roles; KYC documents held by provider, not by the platform
Backups	3-2-1 strategy with immutable off-site copy; scheduled restore drills (see Section 17)
Monitoring	Health endpoints, audit-chain and trigger checks, failed sign-ins, backup age, disk and scheduler checks, with alerting (see Section 17)
Vulnerability management	Dependency and image scanning; static analysis (PHP and Solidity); external penetration test and smart-contract audit [firms to be appointed by ReserveChain]
Incident response	Severity model; maintenance mode; contract pause via multisig; evidence preservation against on-chain anchors
Keys	Hardware-backed multisig for contract administration; low-privilege anchor key; deployer admin renounced after deployment

No security control eliminates risk. External audits and penetration tests have **not yet been performed**.

12. Compliance

12.1 Eligibility controls

Any future participation would require, at minimum: identity verification (KYC) for individuals, business verification (KYB) for entities including beneficial owners, AML risk assessment, sanctions and politically-exposed-person screening, and jurisdictional eligibility. The provider is **[To be decided by ReserveChain]**. The platform currently stores only **status values** (not started, pending, approved, rejected) per member; documents remain with the provider.

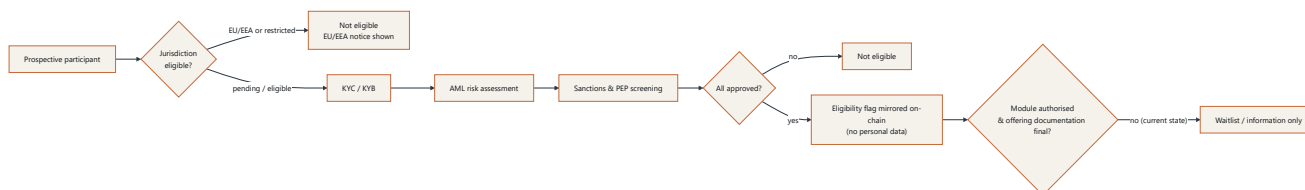


Figure 11 — Compliance gating

12.2 Jurisdictions

- **EU/EEA:** ReserveChain does not currently intend to offer tokens to residents of, or persons located in, the EU/EEA. EU/EEA registrations of interest are recorded as *restricted* and receive general updates only if they opt in.
- **Other jurisdictions:** eligibility is **Pending — subject to legal analysis**. Sanctioned jurisdictions would be blocked.

12.3 Registration of interest

The waitlist collects name, email, country, entity type, optional organisation, program interest and language, together with explicit consent to the disclosure and privacy notice. The version and SHA-256 hash of the consent text are recorded. Double opt-in confirmation is required. **Registration of interest creates no entitlement of any kind** (see Important Notice).

12.4 Data protection

The platform is designed for data minimisation: hashed identifiers in logs, restricted access to personal data, export and erasure procedures, and retention periods **[To be defined by ReserveChain counsel]** under applicable law (including the Swiss Federal Act on Data Protection and, where applicable, the GDPR).

12.5 Communication standards

Public materials use the terms *proposed*, *planned*, *in development* and *subject to final approval*. They do not describe any arrangement as confirmed unless it has been verified through the four-eyes process with evidence, and they do not use language of investment, profit or return.

13. Redemption (Proposed Process)

Redemption is a **proposed** feature. It is **disabled** in the platform and in the RedemptionManager contract. No redemption right exists. Thresholds, fees, delivery terms, eligible locations and timelines are **To be determined — subject to written approval**.

13.1 On-chain process (testnet contract)

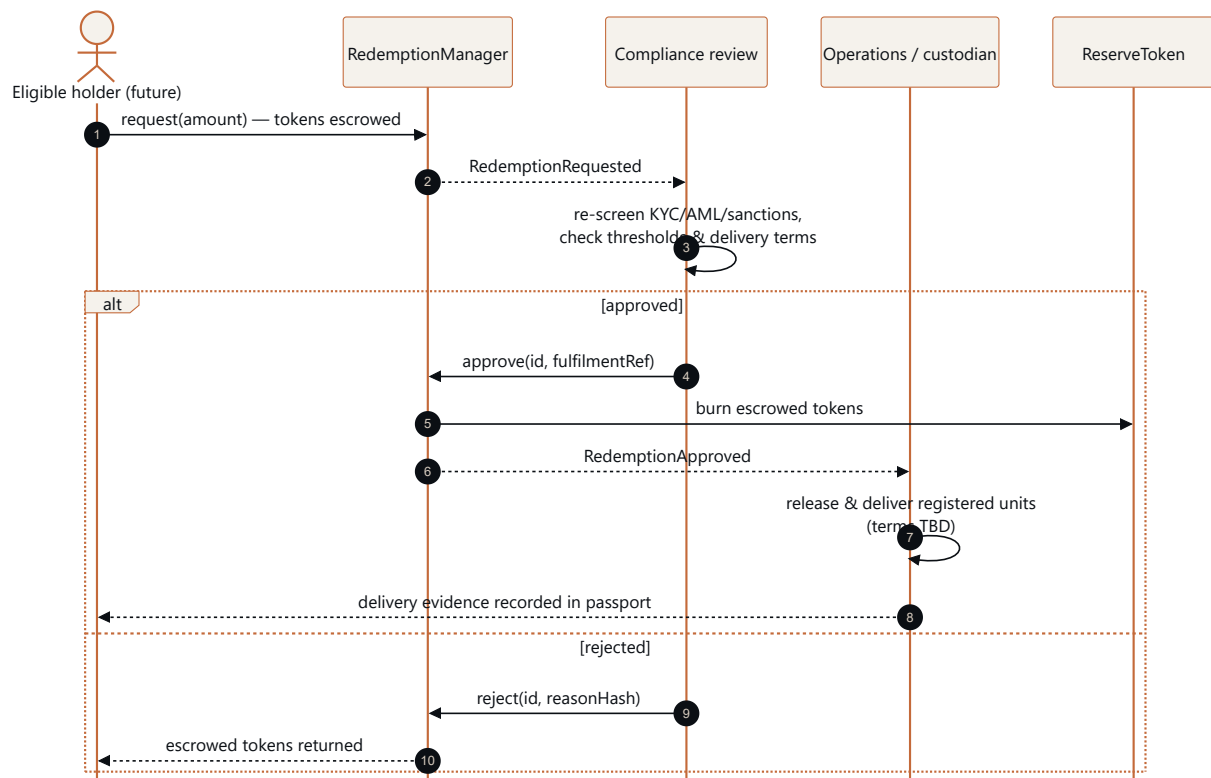


Figure 12 — Redemption flow

Design properties of the proposed process: tokens are escrowed (not burned) until approval, so a rejection is fully reversible; approvals and rejections are recorded on-chain with a fulfilment reference or reason hash; the corresponding CMS redemption record mirrors the state; physical release would be evidenced by custody records attached to the affected passports.

13.2 Off-chain redemption workflow (in development, module inactive)

The platform contains an off-chain workflow that mirrors the RedemptionManager contract. It is part of the gated `redemption` module, which is **inactive** until written authorization. While the module is off, staff can create only clearly labelled **test** redemptions in the development environment.

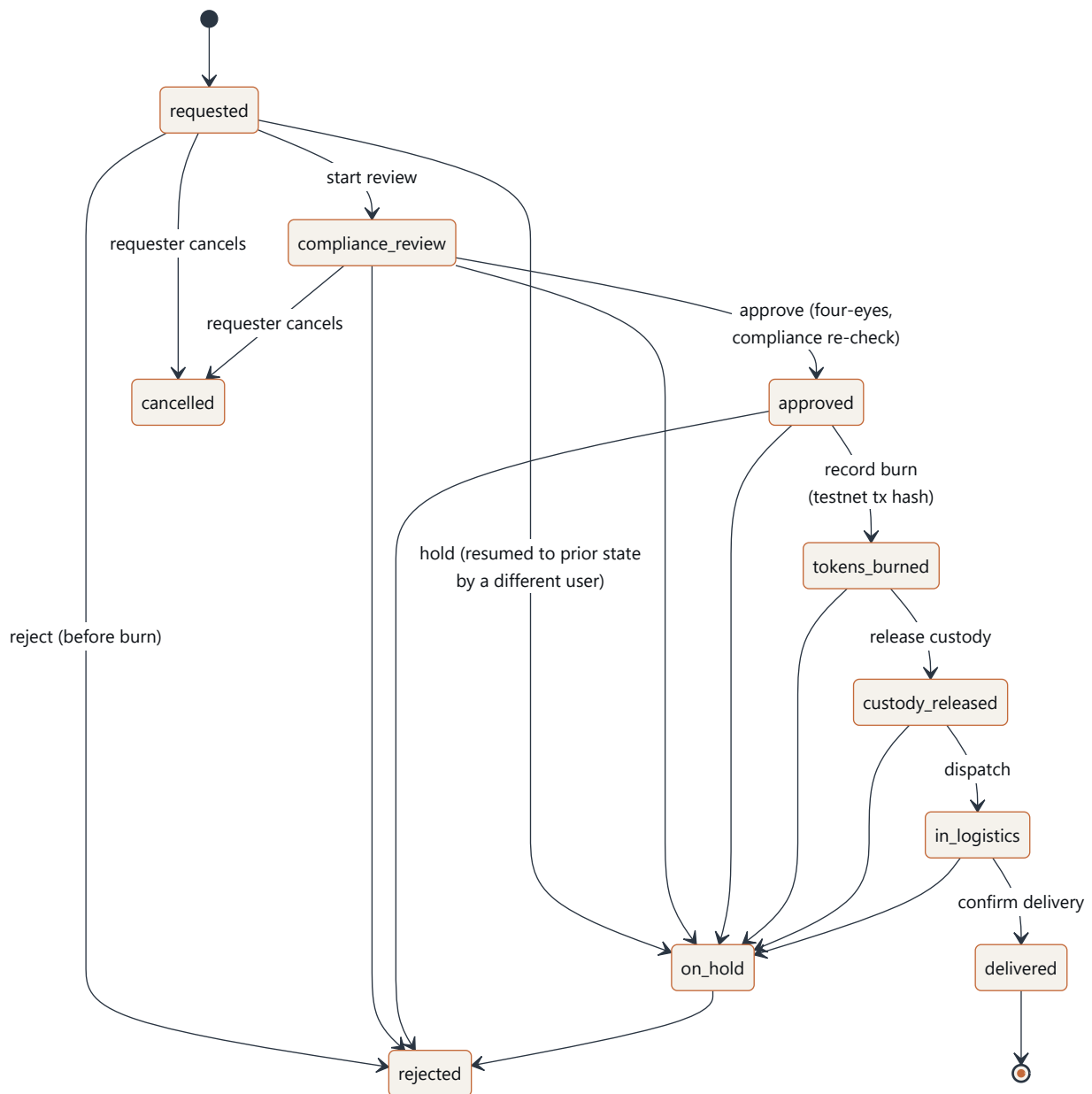


Figure 13 — Redemption states

Control	Implementation (in development)
States	requested → compliance review → approved → tokens burned → custody released → in logistics / customs → delivered; side exits: rejected (staff, before burn), cancelled (requester, before approval), on hold (staff, resumable to the prior state)
Reversibility	Rejection and cancellation are possible only before the burn is recorded, matching the on-chain escrow model; the burn step must carry the test-network burn transaction hash as evidence
Four-eyes	Every transition is capability-checked; the approver must differ from the requester and the preparing operator; the burn recorder must differ from the approver; the delivery confirmer must differ from the dispatcher; a hold must be released by a different user from the one who placed it
Approval binding	Approval is bound to a fingerprint of what was approved (units, lot, amount, program, delivery method, delivery-address hash); a later change invalidates the approval

Control	Implementation (in development)
Compliance re-checks	Eligibility (jurisdiction plus KYC/KYB, AML and sanctions status) is snapshotted at request and re-checked at approval; a restricted jurisdiction or any required check not approved blocks the step and is recorded in the audit trail
Unit-level status	Selected containers, coils or lot are locked against double selection; each unit's redemption status follows the workflow (requested → released → delivered) and is restored on rejection or cancellation, with audit diffs
Thresholds and fees	Never invented: read from the token program or platform settings; when missing they are displayed as <i>to be determined</i> and a live (non-test) request is refused
Personal data	The delivery address is encrypted at rest; only its SHA-256 hash enters the audit trail

The reconciliation engine (Section 14) flags any unit that is accepted into reserve while a redemption is open, to prevent double counting.

14. Proof of Reserves (Proposed)

Proof of Reserves is a **proposed** feature and is **inactive**. No reserve attestation has been published. No attester has been appointed.

The proposed approach combines four elements:

1. **Unit-level registry** — reserves are identified as specific registered units with passports, not as an aggregate figure alone.
2. **Independent attestation** — an attester **[To be appointed by ReserveChain]** issues periodic reports covering the registered units; each report is fingerprinted and registered.
3. **On-chain posting** — the attester posts the attested quantity, the report hash and its URI to ReserveGuard; the mint ceiling is derived from the most recent fresh attestation.
4. **Public reconciliation** — a reserves page would display, per program, the attested units, report date, attester, report fingerprint, on-chain transaction, and tokens outstanding (currently: *Not applicable — no tokens issued*).

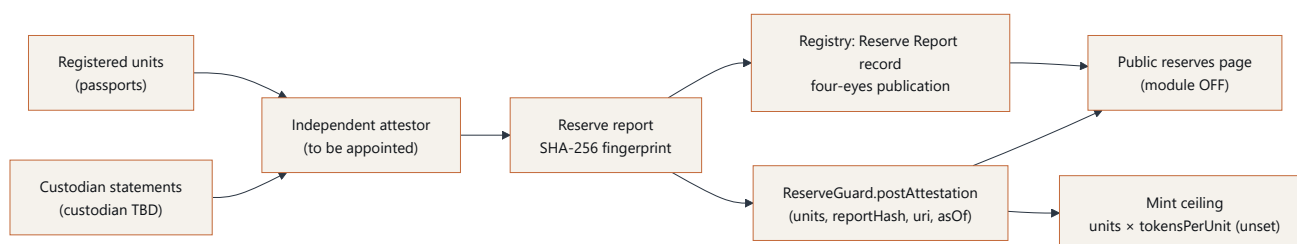


Figure 14 — Proof of reserves

14.1 Reconciliation engine (in development)

For each program, the platform computes a deterministic reconciliation from the registry: registered units by type; declared versus verified weight; custody evidence; reserve eligibility; token supply (read from a test-network contract only when one is configured — otherwise shown as *not deployed* or *RPC not configured*); the latest attestation; and coverage. **Coverage is computed only** from a verified attestation, an approved asset-to-token ratio and a known token supply; otherwise it is shown as *not computed*. No figure is estimated or filled in. Uploaded information is never described as "independently verified" unless the corresponding reserve report itself has been approved with status *verified*.

14.2 Exception rules

Exceptions surface problems instead of smoothing them away. Each carries a severity (critical, warning, information) and a reference to the affected unit, document or program.

Exception	Severity	Condition
Reserve without custody	Critical	Unit marked reserve-eligible but custody not evidenced

Exception	Severity	Condition
Unfingerprinted document	Critical	Linked document has no SHA-256 fingerprint
Insurance expired	Critical	Linked insurance record has expired
Supply without attestation	Critical	Tokens outstanding but no verified attestation exists
Supply exceeds reserve	Critical	Token supply exceeds the attested reserve allowance
No CoA	Warning	No Certificate of Analysis linked to the lot
Missing weight	Warning	Net weight not recorded
Redemption overlap	Warning	Unit accepted into reserve while a redemption is open (double-counting risk)
Insurance expiring	Warning	Linked insurance record approaching expiry
Stale valuation	Warning	Valuation missing or older than 180 days for a reserve unit
Stale attestation	Warning	Latest attestation older than 90 days or undated
Owner-supplied only	Information	Certificate is owner-supplied and not independently verified
No attestation	Information	No reserve attestation published for the program

A daily job records an audit entry and alerts compliance staff whenever the set of open exceptions changes.

14.3 Reserve snapshots

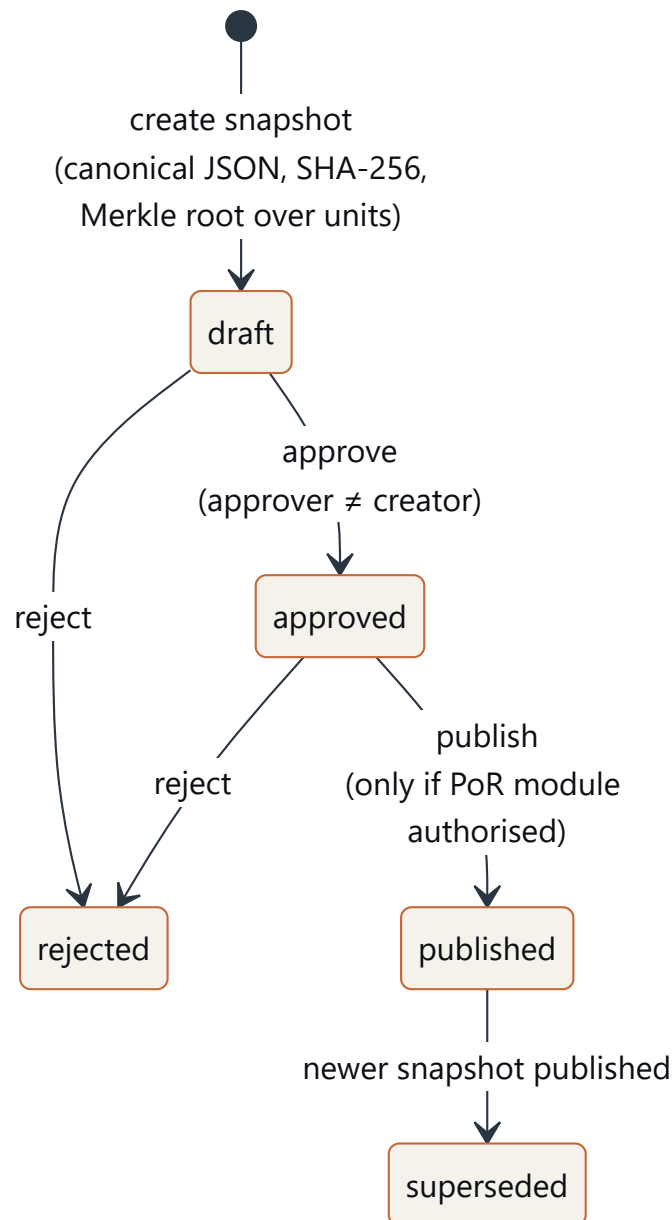


Figure 15 — PoR snapshot lifecycle

A snapshot freezes a reconciliation as canonical JSON, records its SHA-256 hash and a Merkle root over the unit fingerprints, and counts its open exceptions. Snapshots are created manually or as a weekly automatic draft for review. Under the four-eyes rule the creator of a snapshot cannot approve it; only an approved snapshot can be published, and **publication is locked until the Proof of Reserves module is authorised**. Each published snapshot supersedes the previous one for its program, and its integrity can be re-checked against the stored hash. Every step is recorded in the audit trail together with the snapshot hash, so published reports are anchored whenever the audit chain is anchored on-chain. Attestation reports can be uploaded, fingerprinted and routed as draft reserve-report records to the review queue.

Limitations: an on-chain attestation records a statement by a key holder; its reliability depends entirely on the independence, competence and procedures of the attester and the custodian. A reserve report

at a point in time does not guarantee reserves at other times. A reconciliation is only as complete as the registry data behind it.

15. Wallet Linking and Payment Verification (Inactive Modules)

The modules in this section are implemented as source code but are **inactive**. They operate only against **test networks**, and they cannot be activated without written authorization. No purchase, payment or token acquisition is possible through the platform. Nothing in this section is an offer or an invitation to pay.

15.1 Wallet linking (module `wallet`)

A participant could link up to three wallets to an account by signing a Sign-In-with-Ethereum-style (EIP-4361) challenge with `personal_sign` (EIP-191). The signed statement reads *"Link this wallet to your ReserveChain account. No transaction, no fees."* Challenges expire after ten minutes. Signatures are verified **server-side**: the message is rebuilt from the stored challenge, hashed (Keccak-256) and the signer recovered with secp256k1 in pure PHP; where the required maths library is unavailable, the test-network ecrecover precompile is used instead. Linking a wallet creates no entitlement of any kind.

15.2 USDT payment verification (modules `purchase` and `usdt_payments`)

This module would verify ERC-20 USDT transfers. It is gated by **both** the `purchase` and `usdt_payments` modules **and** by a site mode that permits participation, none of which is enabled. There is **no pricing logic** anywhere in the platform: any amount would be entered and approved by staff under four-eyes control, bound to a fingerprint. Verification would require a successful transaction receipt containing a Transfer event from the configured USDT contract, from the participant's linked wallet, to the configured treasury, for exactly the approved amount, with the configured number of confirmations; a scheduled job re-checks pending payment intents (states: created, awaiting transaction, confirming, confirmed, failed, expired).

15.3 Test networks only

Every RPC call first checks the network chain ID against an allow-list of test networks (Ethereum Sepolia, Polygon Amoy, local development) **and** against the platform network setting. Mainnet calls are refused.

16. Participant Portal, Data Rooms and Asset-Owner Intake

16.1 Participant portal (in development)

The participant portal is a single-page interface on top of the platform API, offering sign-in with multi-factor authentication and a personal dashboard. It is the staged-access entry point; registration follows the platform's registration settings. Use of the portal creates no entitlement and is not a channel for any offering.

16.2 Restricted documents and data rooms

Each evidence document carries an **audience**: public, investor, enterprise, auditor, custodian or staff. Restricted files are moved to private storage that denies direct web access and uses unguessable file names; they are only ever streamed through the application via a **signed link that expires after ten minutes** and re-checks access on use. Read-only participant roles (investor data room, enterprise client, custodian) have no administrative access. The investor and enterprise data rooms are **gated** by the `investor_portal` and `enterprise_portal` modules, which are inactive; until they are authorised, non-staff members cannot see documents in those audiences.

16.3 Asset-owner / originator intake (module `asset_owner_portal`)

Producers, owners, originators and custodians of copper powder or nickel wire could submit material for consideration through an intake form gated by the `asset_owner_portal` module. Submissions become private records that move through the pipeline **received** → **initial review** → **due diligence** → **accepted for onboarding** | **declined**. Final decisions are four-eyes: one manager proposes and a different authorised user confirms, bound to a fingerprint of the submission as proposed. Supporting certificates (PDF or images, up to five files of 20 MB) are stored privately, SHA-256 fingerprinted and accessible to staff only. **A submission is not an acceptance, a valuation or a commitment by either side.**

17. Operations: Monitoring, Backups and Disaster Recovery

Operational targets in this section are **proposed** and subject to final approval by ReserveChain; they are not commitments of any hosting provider.

17.1 Monitoring and alerting (implemented for the single-server deployment)

Element	Description
Public health endpoint	GET /wp-json/ic/v1/health — rate-limited, no secrets; reports overall status (ok / degraded), audit chain head and last verification, trigger presence, cron age, disk headroom and backup status
Detailed health endpoint	Token-protected; returns every check with detail, versions, last backup record, recent failures and alert state
Scheduled checks	Database reachability; audit-chain verification; presence of the audit-immutability triggers; failed sign-ins (threshold 20 per hour); waitlist / contact / mail failures; age of last successful backup (26-hour limit); disk usage (85 % threshold); scheduler heartbeat
Alerting	E-mail and optional chat webhook, de-duplicated (repeat interval six hours); an operations widget on the staff dashboard

17.2 Backups (proposed targets)

The proposed backup strategy follows a **3-2-1** model with one off-site, immutable copy held under a separate account and provider whose credentials are not available to the production application. Proposed schedule: continuous point-in-time recovery logs, daily database snapshots, daily encrypted logical dumps off-site, versioned object storage and a daily export of the audit log to write-once storage. Retention periods are subject to counsel's retention policy.

Metric	Proposed target
RPO (data loss)	≤ 15 minutes (point-in-time recovery)
RTO (service restore, same region)	≤ 4 hours
RTO (region loss)	≤ 24 hours

17.3 Restore and disaster-recovery drills

The restore procedure verifies the archive checksum before decryption, refuses to overwrite production without an explicit override, confirms that the audit-immutability triggers are present straight from the dump, never rewrites the audit tables, and succeeds only if the tamper test, a full audit-chain verification, the home page and the health endpoint all pass. A local restore drill was performed on 2026-10-06 against a throwaway environment: the triggers were present after import,

tamper attempts were rejected and the audit chain verified intact. That drill used development data only; drills with production archives are proposed to be repeated quarterly, and a full rebuild in a disaster-recovery region annually and before launch. External assurance of these procedures has **not yet been performed**.

18. Roadmap

The roadmap is **phase-based**. No dates are given; timing depends on legal, regulatory, commercial and technical dependencies, many outside ReserveChain's control. Completion of any phase does not imply that later phases will occur.

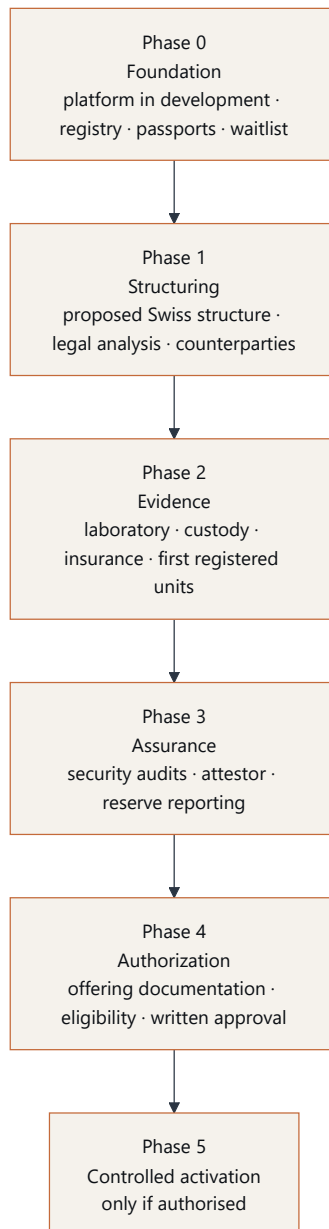


Figure 16 — Roadmap

Phase	Objectives	Status
0 — Foundation	Website, CMS, registry, Digital Asset Passports, audit trail, waitlist, testnet contracts, mobile app source, documentation; reserve reconciliation and snapshots, redemption workflow, wallet and payment-verification modules (inactive), participant portal and data rooms, asset-owner intake, operations monitoring and backups; registration of owner-supplied certificates (pending verification)	In development
1 — Structuring	Corporate and legal structure; regulatory analysis; selection of providers	Proposed
2 — Evidence	Laboratory, custody and insurance arrangements; first units registered with evidence	Proposed
3 — Assurance	Penetration test, smart-contract audit, attestor appointment, first reserve report	Proposed
4 — Authorization	Definitive offering documentation; eligibility framework; written approvals	Proposed
5 — Controlled activation	Activation of gated modules only if all prior conditions are met	Proposed — subject to final approval

19. Risk Factors

The following non-exhaustive risks apply to the project and to any future representation of physical metals. They should be read together with the Important Notice.

19.1 Project and structural risks

- **Development risk.** The platform is in development; features may change, be delayed or never be completed.
- **Entity risk.** The proposed Swiss structure has not been formed or authorised; it may not be formed, or may be formed in a different form or jurisdiction.
- **Counterparty-selection risk.** Laboratories, custodians, insurers, attestors and service providers have not been appointed; suitable counterparties may not be available on acceptable terms.
- **Funding and continuity risk.** The project may lack resources to complete development or operate the platform.
- **Key-person risk.** The project may depend on a small number of individuals.

19.2 Legal and regulatory risks

- **Characterisation risk.** Tokens may be characterised as securities, financial instruments, deposits, commodities derivatives or other regulated products, with licensing, prospectus and conduct consequences.
- **Regulatory change.** Laws on crypto-assets, tokenised securities, AML, sanctions and data protection are evolving and may restrict or prohibit the proposed activities.
- **Jurisdictional restrictions.** Persons in the EU/EEA and potentially other jurisdictions will not be eligible; eligibility rules may change.
- **Enforceability.** Holder rights (if any) may be uncertain or unenforceable in some jurisdictions or in insolvency.
- **Sanctions and AML.** Counterparties, sourcing or participants could become subject to sanctions, triggering freezes or exclusions.

19.3 Asset and custody risks

- **Specification risk.** Material may not meet target specifications; laboratory results may vary between methods and laboratories.
- **Owner-supplied evidence risk.** The certificates currently registered were supplied by the asset owner and have not been independently confirmed with the issuing laboratory; they may prove inaccurate, incomplete, outdated or not attributable to the material. Declared quantities are customer declarations, not verified weights.
- **Degradation risk.** Fine metal powders can oxidise or absorb moisture; wire can be damaged or corrode; storage conditions matter.
- **Custody risk.** Loss, theft, fraud, misallocation, commingling, or custodian insolvency.
- **Insurance risk.** Insurance may be unavailable, insufficient, subject to exclusions or not pay out.

- **Valuation risk.** Industrial metals are subject to price volatility; specialised forms may have thin markets and wide differences between valuation and realisable value. No valuation is published.
- **Provenance and ESG risk.** Sourcing may be subject to responsible-sourcing, environmental and human-rights scrutiny; documentation may be incomplete.
- **Logistics risk.** Transport, export controls, customs, hazardous-material handling and delivery may delay or prevent physical movements.

19.4 Reserve and redemption risks

- **Attestation risk.** Attestations are statements at a point in time and depend on the attestor's procedures and independence.
- **Mismatch risk.** Operational errors could cause registry, reserve reports and on-chain state to diverge. Reconciliation exceptions are designed to surface such divergences but cannot detect errors that are not reflected in the recorded data.
- **Redemption risk.** Redemption may never be available; if available, it may be subject to thresholds, fees, delays, suspension, jurisdiction limits and delivery constraints.

19.5 Technology risks

- **Smart-contract risk.** Contracts may contain vulnerabilities despite testing and use of established libraries; audits have not yet been performed.
- **Key-management risk.** Loss or compromise of administrative or signer keys.
- **Blockchain risk.** Network congestion, forks, reorganisations, fee spikes, client bugs, or changes to the underlying network.
- **Oracle / attestor key risk.** A compromised attestor key could post incorrect data (mitigated, not eliminated, by staleness windows, caps and multisig oversight).
- **Wallet and payment risk.** If ever activated, wallet linking and payment verification would depend on correct signature verification, RPC providers and stablecoin contracts; payments sent to wrong addresses, in wrong amounts or on wrong networks may be unrecoverable. These modules are currently inactive.
- **Platform risk.** Web, CMS, API or app vulnerabilities; third-party dependency compromise; outages.
- **Data risk.** Data loss or corruption despite backups; restoration may lose recent data.

19.6 Market and liquidity risks

- **No liquidity.** No secondary market or liquidity arrangement exists or is represented.
- **Price risk.** Any token, if issued, could trade (where permitted) at prices different from any reference value of the underlying material.
- **Concentration risk.** Two programs only; exposure to two metals and to specific forms.

19.7 Operational and reputational risks

- **Human error** in data entry or approvals, mitigated but not eliminated by four-eyes controls.
- **Fraud by insiders or counterparties.**

- **Misinformation.** Third parties may misrepresent the project or impersonate it; only official channels **[To be provided by ReserveChain]** should be relied upon.

20. Glossary

Term	Definition
AML	Anti-money laundering controls and procedures
Attestation	A statement by an independent party about the existence and quantity of reserves at a point in time
Audit trail	Append-only record of material platform actions, hash-chained for tamper evidence
Chain head	The hash of the most recent audit-trail entry together with its sequence number
Certificate of Analysis (CoA)	Laboratory document reporting the measured composition/properties of a sample
Claim Status	Visible status of a factual claim: proposed, in development, pending verification, verified, not applicable
Coil / spool	A wound unit of wire tracked individually in the registry
Custody record	Evidence of receipt, holding, transfer or release of a unit by a custodian
Data room	Audience-restricted document area served only through signed, expiring links; investor and enterprise data rooms are gated modules
DAP — Digital Asset Passport	Public, per-unit page showing identity, fields with status, timeline, evidence ledger and Merkle root
EEA	European Economic Area (EU member states plus Iceland, Liechtenstein, Norway)
ERC-20	A standard interface for fungible tokens on Ethereum-compatible blockchains
Fail-closed	A design in which a control defaults to the most restrictive state when not fully configured
Four-eyes principle	Requirement that a second, different authorised person approves a change before it takes effect
KYB	Know Your Business — verification of a legal entity and its beneficial owners
KYC	Know Your Customer — verification of an individual's identity
Merkle root	A single hash committing to a set of hashes, enabling inclusion proofs
Multisig	A wallet requiring signatures from multiple keys to execute a transaction
Owner-supplied	Provenance label for evidence provided by the asset owner and not yet independently confirmed with its issuer
PEP	Politically exposed person
Proof of Reserves (PoR)	A process for demonstrating that represented assets exist, proposed here as a combination of registry, attestation and on-chain posting

Term	Definition
Reconciliation exception	A rule-based flag raised when registry, custody, insurance, attestation or supply data are missing, stale or inconsistent
Reserve snapshot	A fingerprinted, four-eyes-approved record of a program's reconciliation at a point in time
ReserveGuard	Contract computing a mint ceiling from attested units and a configured ratio
SHA-256	Cryptographic hash function producing a 256-bit fingerprint
SIWE	Sign-In with Ethereum (EIP-4361): a signed text challenge proving control of a wallet without a transaction
Testnet	A public blockchain network used for testing, whose tokens have no economic value
USDT	A US-dollar stablecoin issued by a third party as an ERC-20 token; referenced only by the inactive payment-verification module
Waitlist	Registration of interest; creates no entitlement

21. Document Control

Item	Value
Document	ReserveChain Institutional Whitepaper — Discussion Draft
Version	Draft v0.10
Status	In development — subject to legal review and final approval
Owner	ReserveChain [responsible person to be provided]
Approval	Pending — subject to final approval
Distribution	Information only; not an offering document
Diagram sources	docs/whitepaper/diagrams/*.mmd
Build	docs/whitepaper/build/ (Markdown → DOCX and PDF)
Changes in Draft v0.10	Owner-supplied certificates (Section 4.3); off-chain redemption workflow (Section 13.2); reserve reconciliation, exceptions and snapshots (Sections 14.1–14.3); new Sections 15–17; risk factors and glossary updated

Mandatory disclosure. ReserveChain is currently in development. No tokens are being offered or sold through this website. Registration of interest does not constitute an investment, token purchase, asset reservation, price reservation, token allocation or entitlement to participate in any future offering. Any future availability will be subject to the final Swiss corporate and legal structure, definitive offering documentation, asset verification, custody arrangements, jurisdictional eligibility, KYC/KYB, sanctions screening and final approval.

EU/EEA notice. ReserveChain does not currently intend to offer tokens to residents of, or persons located in, the EU/EEA.